

Empowering Visually Impaired Using Machine Learning

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Abstract—This study introduces a compact assistance system tailored for people with visual impairments. The application combines three essential features: banknote identification, sign language translation, and converting images to spoken text. Our approach utilizes lightweight machine learning models optimized for resource constrained devices. We use a Convolutional Neural Network paired with Support Vector Machine and Random Forest algorithms to recognize currency. For sign language translation, MediaPipe extracts hand landmarks which are then classified using standard machine learning techniques. The optical character recognition component leverages Tesseract, enhanced by n gram models and custom rule based corrections to ensure text clarity. Testing indicates robust accuracy across these functions, proving that an integrated, fully offline, and real time application is highly viable.

Index Terms—assistive technology, machine learning, visual impairment, currency recognition, sign language translation, optical character recognition, text to speech.

I. INTRODUCTION

People with visual impairments regularly face difficulties with routine activities, such as distinguishing paper money, reading physical documents, and communicating with those using sign language. Even though machine learning has made great strides in solving these individual problems, current applications are usually limited to single tasks or demand significant computing power. This initiative aims to consolidate these three capabilities into one efficient application that runs entirely offline.

Novelty of this work:

- An all in one tool offering currency detection, gesture translation, and text reading
- Efficient models that run smoothly on basic mobile phones
- Completely offline processing to protect user privacy
- Optical character recognition improvements using n gram and custom logic
- A newly compiled dataset specifically for Bangladeshi banknotes

II. MOTIVATION BEHIND THE WORK

Research Context: Current assistive tools usually target only one problem or require continuous internet access. However, users need diverse tools in a single package, and offline functionality is critical for privacy and consistent access.

Practical Applications:

- Identifying money to prevent fraud and financial mistakes
- Translating gestures to engage with mute individuals
- Reading text aloud for medication labels, receipts, and important documents

III. RELATED WORK

Previous research on currency classification heavily utilizes deep network architectures like MobileNetV2 and ResNet [1]. Translating hand gestures frequently involves MediaPipe or similar frameworks paired with standard classification models [2], [4]. Systems focused on reading text usually deploy Tesseract without refining the final output [3]. Other studies have built smart backpacks with currency detection, obstacle sensors, and GPS tracking to promote daily independence [8]. While these individual projects are successful, they fail to bundle these features into a single offline mobile platform, which is exactly what our research accomplishes.

IV. PROPOSED METHODOLOGY

A. Currency Detection

- Images are resized and normalized for processing
- A compact Convolutional Neural Network extracts key features
- A Support Vector Machine or Random Forest model handles the final classification
- A confidence threshold filters out uncertain predictions

B. Sign Language to Speech

- MediaPipe identifies 21 specific hand coordinates
- These points are processed by a standard classifier
- The resulting gesture is then translated into audible speech

C. Text to Speech

- Pytesseract captures the initial raw text
- N gram ranking helps correct uncertain words [5]
- Custom logic rules fix standard recognition errors
- An integrated audio module then reads the finalized text [6]

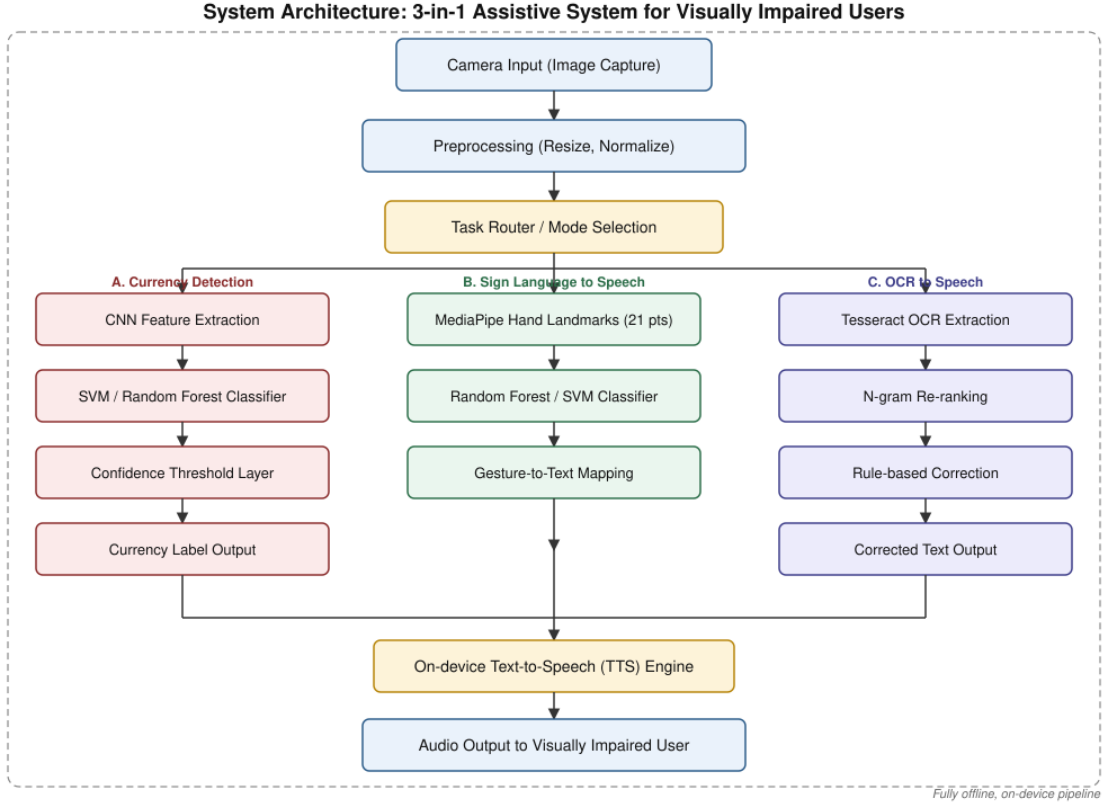


Fig. 1. Proposed system architecture combining currency detection, sign language recognition, and text reading into a single offline pipeline.

D. System Architecture

Fig. 1 illustrates how these components operate within a unified workflow. When an image is captured, it undergoes preprocessing and is routed to the appropriate module. Each path performs its unique feature extraction and classification. Finally, the results merge at a common text to speech component built into the device. Handling everything locally eliminates internet delays and secures the user's data.

V. EVALUATION AND FINDINGS

We tested the pipeline using three datasets:

- A custom collection of Bangladeshi currency containing 3600 training images and 900 validation images
- An Indian Sign Language dataset obtained from Kaggle [7]
- Various mixed document images for text testing

Training Performance:

- The currency identifier achieved an accuracy of 92 percent
- The sign language module achieved an accuracy of 86 percent
- Inference speeds averaged 0.29 seconds for currency and 0.35 seconds for gestures

Performance Comparison:

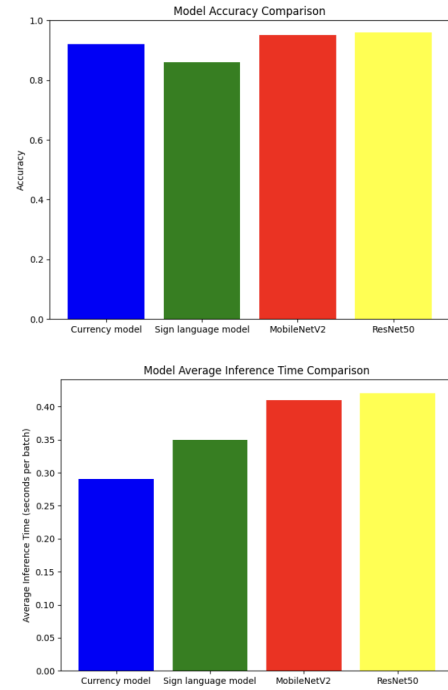


Fig. 2. Model accuracy and average inference time comparison across the currency model, sign language model, and two reference network backbones.

VI. DISCUSSION

In contrast to prior research, our framework prioritizes lightweight models, sacrificing a marginal amount of accuracy to achieve substantial gains in execution speed and device portability. This integrated setup proves that local machine learning can effectively assist visually impaired users without relying on cloud services.

Current Limitations:

- The recognizable gesture library is currently small
- The currency dataset needs broader variations such as damaged or aged bills
- The text reader faces challenges with handwriting and poor lighting
- We have not yet implemented mobile quantization like TFLite

VII. SYSTEM DESIGN AND DEPLOYMENT CONSIDERATIONS

A. Software and Hardware Stack

We built this prototype using accessible open source software to ensure it functions on standard consumer hardware rather than requiring specialized equipment. OpenCV manages image capture and preprocessing, scikit learn handles the machine learning classifications, MediaPipe tracks hand movements, and Pytesseract extracts text. A local text to speech engine vocalizes the output, bypassing internet requirements. This efficient stack allows the system to run on mid tier Android phones or simple single board computers, keeping costs minimal.

B. Latency and Resource Budget

Because the modules operate on the same device, minimizing response time was a priority. We selected classical classifiers for the currency and gesture tasks because they execute in milliseconds, keeping the entire process under half a second. This ensures the user receives prompt audio feedback without frustrating delays.

C. Interaction Design

The interface relies entirely on audio prompts since the target users cannot see a screen. Voice cues confirm the active mode, and all outcomes are spoken rather than displayed visually. Users can navigate the application using physical buttons or brief voice commands, ensuring complete accessibility for individuals with severe vision loss.

VIII. BROADER IMPACT AND ETHICAL CONSIDERATIONS

Operating offline provides a crucial layer of privacy. Sensitive items like bank notes, medical prescriptions, and personal documents are processed entirely on the user's phone, ensuring no private data is uploaded to remote servers. This approach also guarantees functionality in regions with unstable internet connections, which is particularly relevant for users living in Bangladesh.

Furthermore, prioritizing classical machine learning over massive deep learning networks makes the software accessible

on budget friendly smartphones. This significantly reduces financial barriers for users in developing areas. Future field testing with visually impaired individuals will help refine the audio interface and validate its usability in practical scenarios beyond the standard training metrics.

IX. CONCLUSION AND FUTURE WORK

This study showcases a highly practical, offline assistive application that tackles three major daily challenges for visually impaired users. Moving forward, we intend to expand our datasets, introduce full sentence translation for sign language, and improve the ability to read handwritten text. Additionally, we plan to conduct real world usability trials, include more international currencies, and apply model quantization to further minimize battery usage and latency on basic smartphones.

REFERENCES

- [1] H. Hassanpour, A. Yaseri, and G. Ardeshtiri, "Feature extraction for paper currency recognition," in *Proc. 9th Int. Symp. Signal Processing and Its Applications*, 2007, pages 1 to 4.
- [2] K. M. S. V. Praneetha, "Sign Language Detection Using Mediapipe and ML," *IJRASET*, Apr. 2025.
- [3] K. Joshi, "Study of Tesseract OCR," *GLS KALP Journal of Multidisciplinary Studies*, vol. 1, no. 2, pages 41 to 50, Mar. 2024.
- [4] L. Chandwani et al., "Gesture based Sign Language Recognition system using Mediapipe," *Preprint*, Jun. 2023.
- [5] Tong and Evans, "Context Aware OCR Correction using N Gram Language Models," *Pattern Recognition Letters*, 2019.
- [6] Sainath et al., "Text to Speech Synthesis with Deep Neural Networks," *ICASSP*, 2015.
- [7] P. Arikeri, "Indian Sign Language Dataset," *Kaggle*, 2022.
- [8] A. I. Shihab, M. S. H. Imon, T. A. S. Niloy, M. A. Islam, S. Islam, and R. M. Talukder, "Smart Assistance System for the Visually Impaired," in *Proc. 2025 28th Int. Conf. Comput. Inf. Technol. (ICCIT)*, Cox's Bazar, Bangladesh, Dec. 2025.